

# TrES-3b

R-2411

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-3\_120325 · created 2026-08-21T01:39:20+00:00

## BENCHMARK: MARGINAL

### Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2456011.95784 +/- 0.00094 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.196 +/- 0.028                   |
| Transit depth (Rp/Rs)^2                           | 3.83 +/- 1.08 [%]                 |
| Orbital Inclination (inc)                         | 81.57 +/- 1.14                    |
| Airmass coefficient 1 (a1)                        | 1.025 +/- 0.012                   |
| Airmass coefficient 2 (a2)                        | -0.017 +/- 0.026                  |
| Scatter in the residuals of the lightcurve fit is | 1.2 %                             |
| Best Comparison Star                              | #2 - [549, 97]                    |
| Optimal Aperture                                  | 3.04                              |
| Optimal Annulus                                   | 5.56                              |
| Transit Duration (day)                            | 0.054 +/- 0.012                   |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                            |
|---------------------------|--------------------------------------------|
| Rp/R■ fit vs published    | 0.196 ± 0.028 vs 0.16309 (deviation 1.18σ) |
| Duration fit vs published | 0.054 d vs 0.0567 d (deviation 0.22σ)      |
| Mid-time O–C              | -3.2 min                                   |
| Depth SNR                 | 7.0                                        |
| Residual scatter          | 1.2 %                                      |

### Light curve

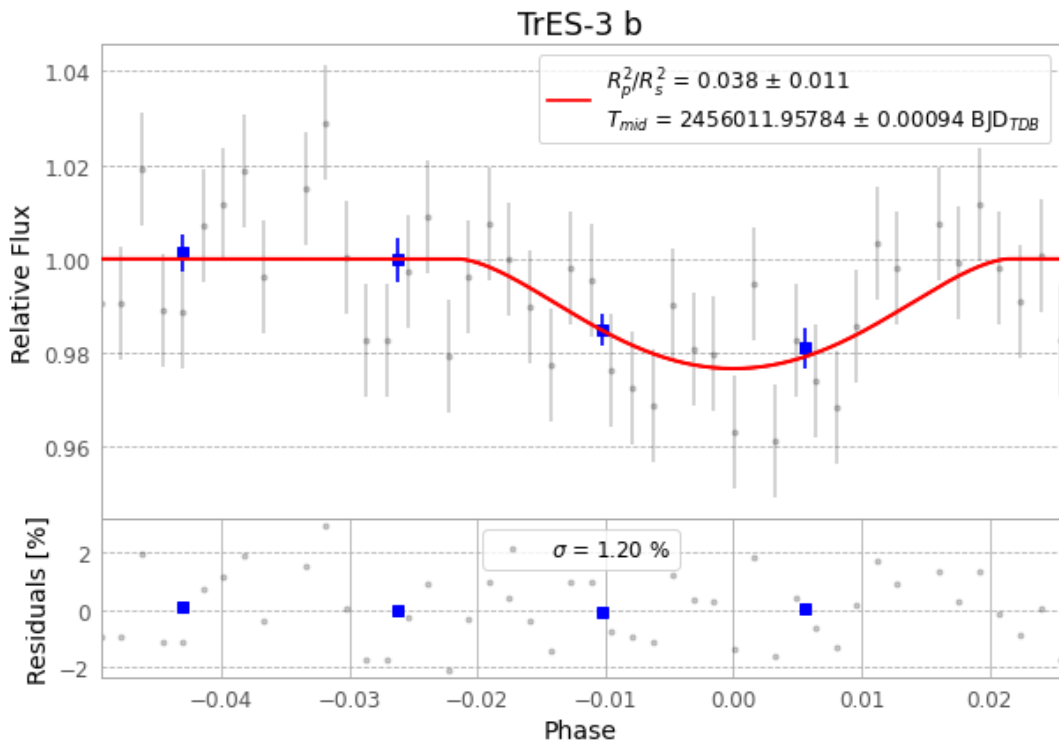


Figure 1 — FinalLightCurve\_TrES-3 b\_2012-03-25.png (SHA-256 389423a4df7342fc...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 206], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 361b9d89255f6608...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

48 FITS frames (31 MB), TRES-3120325092412.FITS → TRES-3120325114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-120325.log (ed4dd15ada2b...), FinalLightCurve\_TrES-3 b\_2012-03-25.png (389423a4df73...), FinalLightCurve\_TrES-3 b\_2012-03-25.csv (e67232010fa3...)

## Compute resources

Wall time 149.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 01:39Z.